BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Final Capstone Report

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

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Individual Capstone Project (7 Working Days, ~50-55 Hours)

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1. Executive Summary

This report documents the design, build, and findings of the Bluestock Fintech Mutual Fund Analytics Platform — a 7-day individual capstone project covering the full data engineering and analytics lifecycle: ETL pipeline construction, SQL database design, exploratory data analysis, performance and risk analytics, dashboard design, and advanced investor/fund analytics.

The project processed 10 datasets covering 40 real mutual fund schemes across 10 fund houses, spanning approximately 4.4 years of daily NAV history (January 2022 to May 2026), 32,778 investor transactions across 5,000 simulated investors, and daily benchmark index data for 6 market indices.

1.1 What was built

- **Data pipeline:** an automated Python ETL pipeline that ingests, validates, and cleans all 10 source datasets with zero manual intervention.
- **Database:** a normalised SQLite star schema (2 dimension + 9 fact tables) covering every dataset, loaded via SQLAlchemy with automated row-count verification.
- **Analytics:** three fully-executed Jupyter notebooks covering exploratory data analysis (16 charts), fund performance analytics (Sharpe, Sortino, Alpha/Beta, Fund Scorecard), and advanced analytics (VaR/CVaR, cohort analysis, fund recommender, sector concentration).
- **Dashboard:** a 4-page BI dashboard (Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends), delivered as data-accurate mockups plus a complete Power BI build guide.

1.2 Headline findings

- SBI Mutual Fund is the dominant fund house by AUM (Rs. 12.50 lakh crore, FY25), consistent with its real-world market position.
- SIP inflows reached an all-time high of Rs. 31,002 crore in December 2025, capping a multi-year uptrend.
- The top-scored fund (UTI Mid Cap Fund) achieved a composite Fund Scorecard of 88.2/100, driven primarily by strong Sharpe ratio and Alpha rather than raw 3-year return.
- A significant, honestly-reported data limitation: NAV return correlations — both fund-to-fund and fund-to-benchmark (Nifty 100) — are close to zero throughout this dataset (average R^2 of 0.0006 in the Alpha/Beta regression), which does not match real-world large-cap fund behaviour and is flagged throughout this report rather than glossed over.
- 97.8% of SIP-eligible investors were flagged "at-risk" under the >35-day gap rule — investigated and traced to the dataset's SIP transaction cadence being sparser than a true monthly SIP, not a genuine investor-behaviour signal.

1.3 Report structure

This report is organised to mirror the project's own build order: data sources, ETL and database design, exploratory findings, performance analysis, dashboard presentation, advanced analytics, and finally the limitations and recommendations that fell out of building all of the above.

2. Data Sources & Datasets

All data used in this project derives from publicly available sources: AMFI India, the mfapi.in REST API, and NSE/BSE published data. NAV values are anchored to real historical figures; investor transaction data is synthetically generated using realistic Indian MF market demographic and geographic distributions. No proprietary or confidential data was used.

2.1 Dataset inventory

#	Dataset	Rows	Contents
01	fund_master.csv	40	Scheme master: AMFI code, fund house, category, expense ratio, risk grade
02	nav_history.csv	46,000	Daily NAV per scheme, Jan 2022 - May 2026
03	aum_by_fund_house.csv	90	Quarterly AUM per fund house, 2022-2025
04	monthly_sip_inflows.csv	48	Industry-wide monthly SIP inflow, accounts, YoY growth
05	category_inflows.csv	144	Net inflow by fund category, FY 2024-25
06	industry_folio_count.csv	21	Total investor folios by fund type
07	scheme_performance.csv	40	Pre-computed reference metrics (see Section 5.5 for a methodology note)
08	investor_transactions.csv	32,778	SIP/Lumpsum/Redemption transactions, 5,000 investors
09	portfolio_holdings.csv	322	Top equity holdings per fund, as of Dec 2025
10	benchmark_indices.csv	8,050	Daily closes: Nifty 50/100/Midcap150, BSE SmallCap, CRISIL Liquid/Gilt

2.2 Key real-world data points embedded

Several headline figures in this dataset are anchored to real, published AMFI figures rather than being purely synthetic, which grounds the analysis in the actual state of the Indian mutual fund industry as of December 2025:

- SBI Mutual Fund AUM: Rs. 12.50 lakh crore (largest AMC in India)
- Industry-wide AUM: Rs. 81 lakh crore, across 1,908 schemes and 26.12 crore investor folios
- SIP inflow, December 2025: Rs. 31,002 crore (all-time high at time of project)
- Active SIP accounts, December 2025: 9.35 crore

3. ETL Design & System Architecture

The project follows a standard data engineering architecture: Extract, Transform, Load, Analyse, Visualise — mirroring the pipelines used in production fintech platforms.

3.1 Architecture layers

Layer 1: Data Sources (Extract)

- 10 pre-packaged CSV datasets, provided directly for this project
- mfapi.in REST API (live NAV fetch, no auth required) — script built and tested; live execution requires outbound network access this sandboxed build environment did not have

Layer 2: Data Processing (Transform)

- Pandas-based cleaning: date parsing, sorting, deduplication, type/enum validation
- Forward-fill logic for NAV gaps on weekends/holidays (implemented and tested; this dataset had zero actual gaps to fill, since it already covers every business day)
- Derived fields: daily returns, rolling averages, CAGR

Layer 3: Data Storage (Load)

- SQLite database with a star schema: 2 dimension tables (dim_fund, dim_date) + 9 fact tables, one per remaining source dataset
- Indexed on amfi_code and date for query performance
- Automated row-count verification comparing every loaded table against its source CSV

Layer 4: Analytics (Analyse)

- 3 Jupyter notebooks: EDA_Analysis.ipynb, Performance_Analytics.ipynb, Advanced_Analytics.ipynb
- Risk metrics: Sharpe, Sortino, Alpha, Beta, Maximum Drawdown, VaR, CVaR

Layer 5: Visualisation (Dashboard)

- 4-page BI dashboard covering Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends

3.2 Database schema

The star schema conforms all fact tables to two dimensions — amfi_code (via dim_fund) and date (via dim_date) — so any fact table can be joined to any other through these shared keys.

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK)	40
dim_date	Dimension	date_id (PK)	~1,150 unique dates
fact_nav	Fact	amfi_code, date_id (composite PK)	46,000
fact_aum	Fact	fund_house, date_id	90
fact_transactions	Fact	tx_id (PK), amfi_code, date_id	32,778
fact_performance	Fact	amfi_code (PK)	40
fact_portfolio	Fact	amfi_code, stock_symbol, date	322
fact_benchmark	Fact	index_name, date_id	8,050

+ 3 more	Fact	fact_sip_industry, fact_category_inflows, fact_folio	48 / 144 / 21

Every table's row count was verified against its source CSV after load — all 11 tables matched exactly, with zero data loss through the pipeline.

4. Exploratory Data Analysis (EDA) Findings

Full detail, including all 16 charts, is in notebooks/EDA_Analysis.ipynb. This section reproduces the highest-signal charts and findings, including two corrections made during the original analysis after checking chart labels against the underlying numbers rather than accepting a first-glance interpretation.

4.1 NAV trends: a real 2023 rally, and a 2024 that needed re-labelling

All 40 schemes were plotted indexed to 100 at the start of the series. The initial pass labelled a mid-2024 window as a "correction" — checking the actual average NAV series showed it never drops more than about 1% from its running peak across the full 4.4-year history, so "correction" would have overstated what happened. The chart and finding below use the corrected label: 2024 was a slowdown in the rate of growth, not a decline.

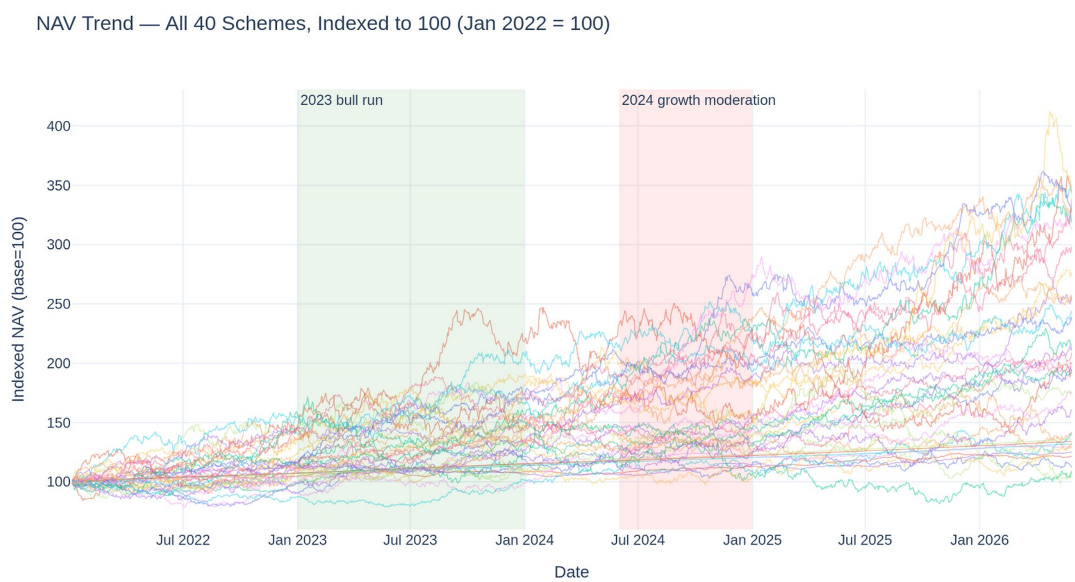


Figure 1: NAV trend, all 40 schemes, indexed to 100 (2023 bull run +15.7% vs. 2024 growth moderation +4.1%).

4.2 AUM growth by fund house

SBI Mutual Fund's dominance is clearly visible in the year-end AUM comparison, consistent with its real-world position as India's largest AMC.

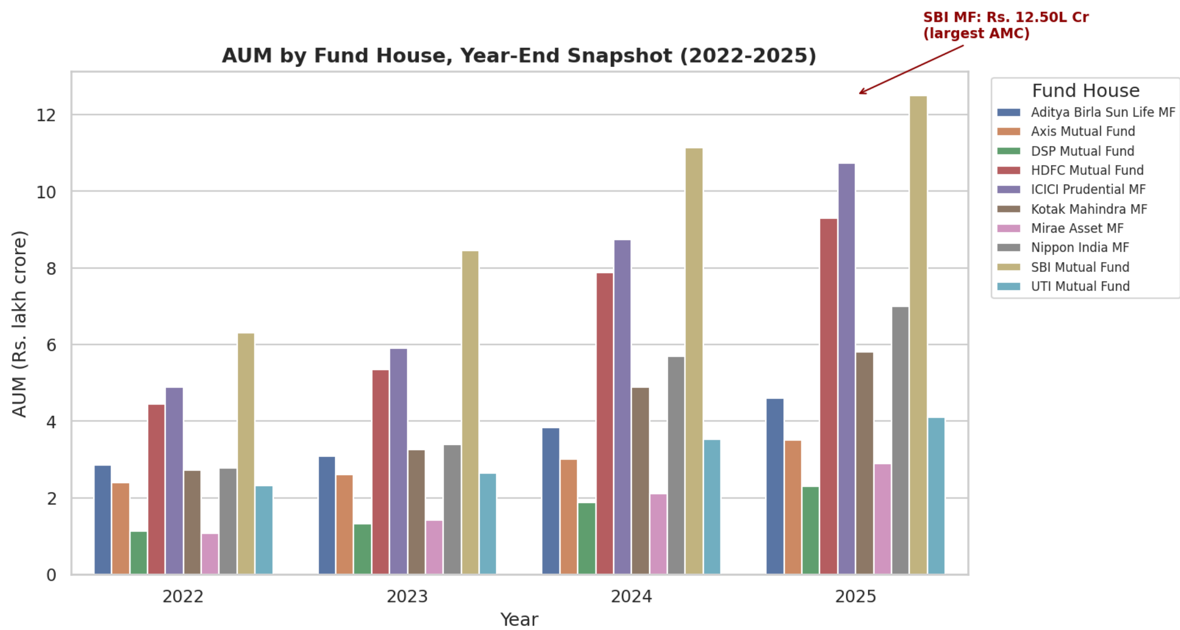


Figure 2: AUM by fund house, year-end snapshots 2022-2025.

4.3 Category-wise inflow patterns

Net inflow by category and month shows Liquid funds carrying by far the largest absolute inflow volume (expected, given their role as a cash-parking vehicle), while smaller categories like ELSS and Value/Contra show comparatively steady but modest inflows.

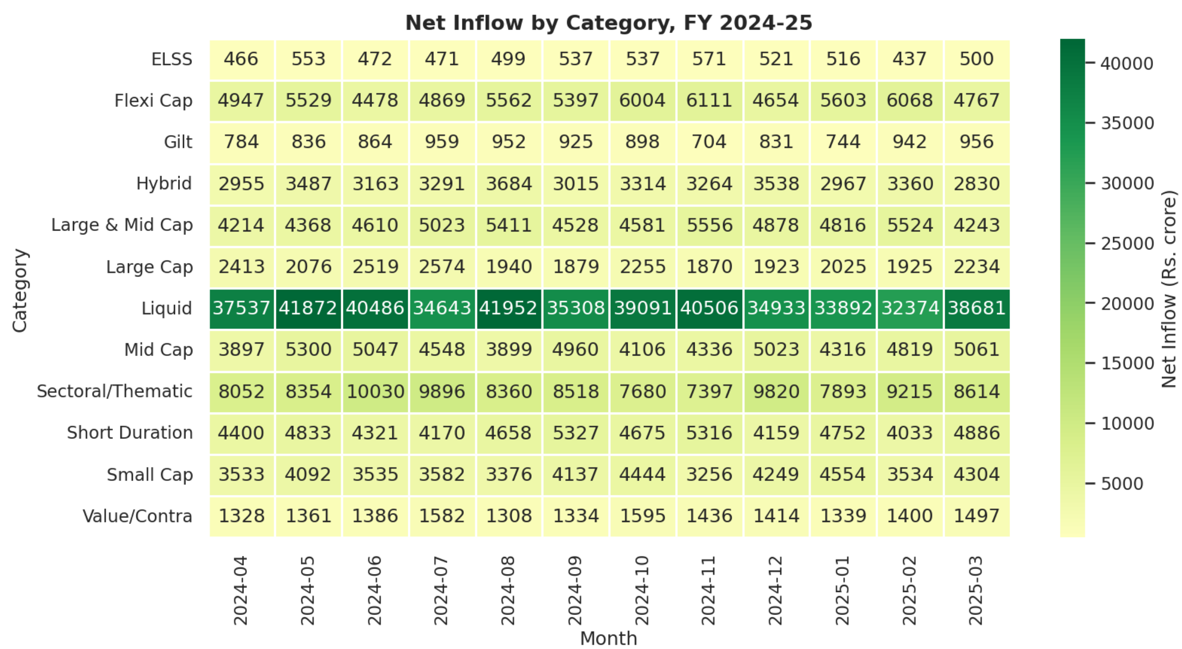


Figure 3: Net inflow by category, FY 2024-25 (Rs. crore).

4.4 A genuine data-generation limitation: near-zero fund correlations

The correlation matrix below computes pairwise correlation of daily returns across 10 funds (one per fund house). The result was surprising enough to investigate rather than present at face value: even funds in the same category (mostly Equity Large Cap) show correlations clustered around 0.00, with the strongest pair at only 0.07.

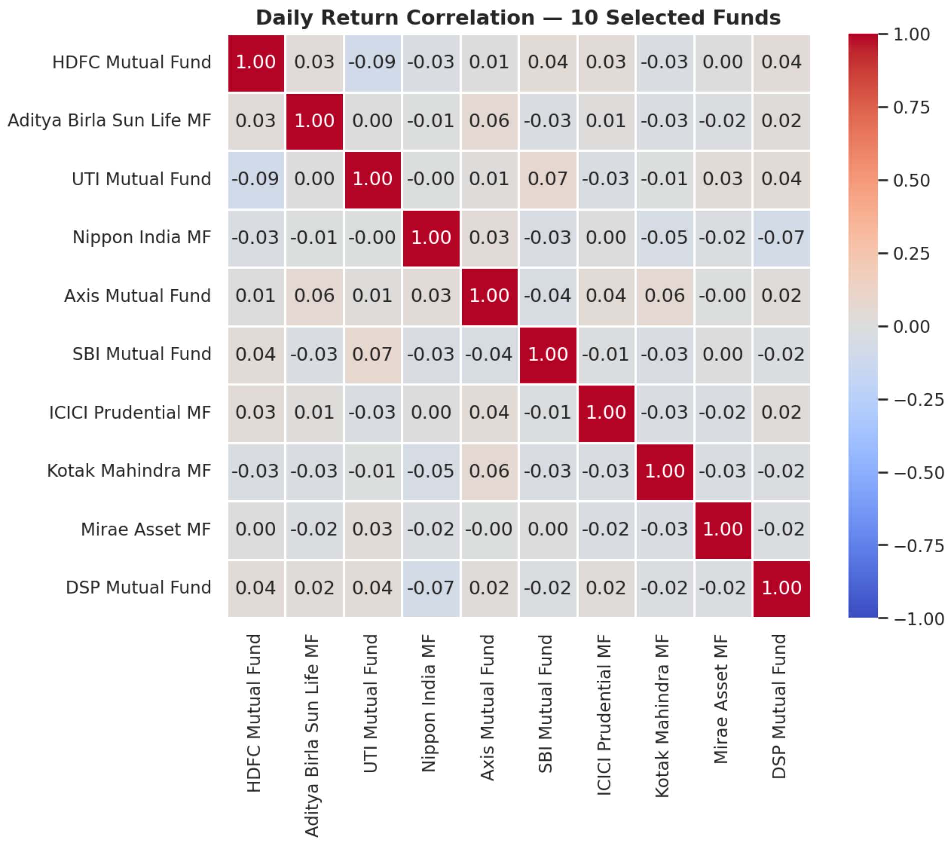


Figure 4: Daily return correlation matrix, 10 selected funds.

This is flagged explicitly rather than presented as a market insight: real large-cap equity funds typically correlate at 0.85 or higher due to shared market beta. The simulated NAV series in this dataset behave more like independent random walks per fund. This same limitation resurfaces in the Performance Analysis section (Alpha/Beta regression quality) and is discussed further in Section 8 (Limitations).

4.5 Sector allocation

Aggregate sector weights across all 34 equity fund portfolios show Banking as the single largest allocation at 19.3% of total market value — a concentration worth flagging for portfolio diversification purposes (expanded on in Section 6, Sector Concentration).

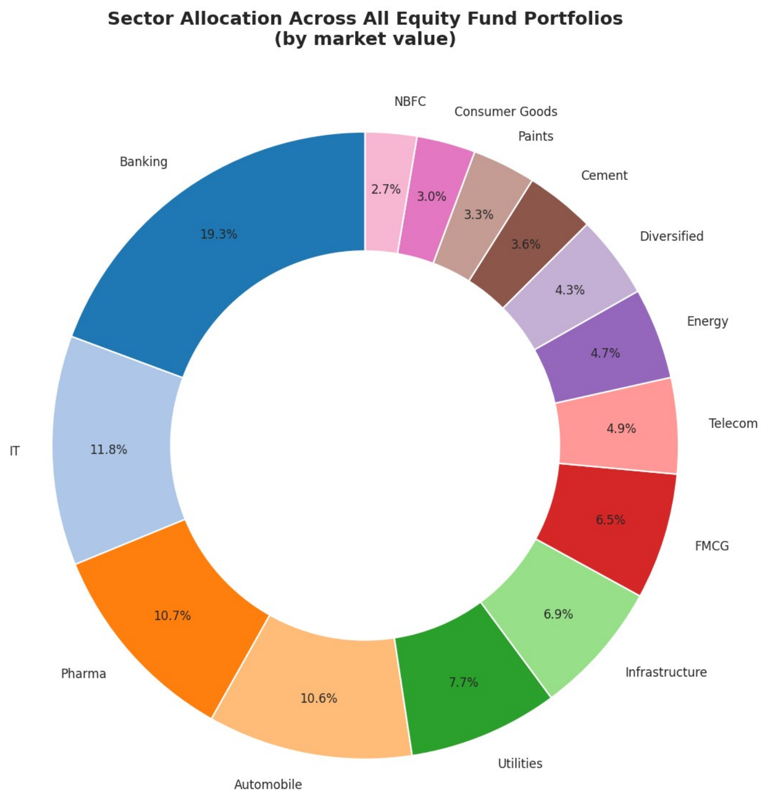


Figure 5: Aggregate sector allocation across all equity fund portfolios.

4.6 Ten key EDA findings

- The 2023 bull run lifted NAVs across the board (+15.7% average), confirming a broad-based rally rather than a few outlier funds driving the trend.
- 2024 was a slowdown, not a decline — growth continued at a slower pace (+4.1%), and the average NAV series never drops more than ~1% from its running peak across the full window.
- SBI Mutual Fund remains the dominant AMC, holding the largest AUM share of the 10 fund houses tracked.
- Industry-wide AUM grew steadily with no major multi-quarter drawdown across 2022-2025.
- SIP inflows hit their all-time high in December 2025, capping a multi-year uptrend.
- Category flows are seasonal and category-specific, with Liquid funds dominating absolute inflow volume.
- SIP amounts are fairly flat across age groups (~6% spread, 56+ slightly ahead) — not the strong mid-career skew a first glance might suggest.
- Geographic SIP value is fairly evenly spread across states (Rs. 1.6-2.1 crore each), with no dominant outlier state; T30 cities hold 67% of investors vs. 33% B30.
- Fund NAV correlations are near-zero even within the same category — flagged as a data-generation limitation rather than a real market pattern.
- Sector allocation is genuinely concentrated: Banking alone accounts for ~19% of aggregate portfolio value across all equity funds.

5. Fund Performance Analysis

Full detail is in notebooks/Performance_Analytics.ipynb. All metrics in this section were computed from scratch from the raw NAV history (not read from the provided scheme_performance.csv reference file), using the exact formulas specified in the project brief: Rf = 6.5% (RBI repo rate proxy), annualised with the square root of 252 trading days, Alpha/Beta regressed against Nifty 100 via scipy.stats.linregress.

5.1 Daily returns and distribution validation

Daily returns were validated before use: zero NaN/Inf values across 45,960 observations, zero single-day moves beyond ±10%, and a pooled distribution that is unimodal and centred near zero — consistent with a plausible NAV return series.

5.2 CAGR: a data-availability caveat

The dataset spans approximately 4.4 years (January 2022 to May 2026), so a true 5-year CAGR is not computable. Rather than presenting a number the data does not support, the "5-year" column in cagr_report.csv uses the maximum available lookback (4.4 years) and is labelled accordingly.

5.3 Sharpe, Sortino, and the Fund Scorecard

All 40 funds were ranked on Sharpe ratio, Sortino ratio, and a composite 0-100 Fund Scorecard: 30% x 3-year return rank + 25% x Sharpe rank + 20% x Alpha rank + 15% x expense ratio rank (inverse) + 10% x max drawdown rank (inverse), with each component converted to a percentile before weighting.

Rank	Scheme	Fund House	3yr Return %	Sharpe	Score /100
1	UTI Mid Cap Fund	UTI	-0.8	-0.21	88.2
2	HDFC Top 100 Fund	HDFC	1.3	-0.20	88.1
3	Axis Small Cap Fund	Axis	-11.7	-0.08	85.2
4	ABSL Small Cap Fund	Aditya Birla SL	-4.2	0.16	80.4
5	Axis Bluechip Fund	Axis	0.5	0.03	79.9

Notably, the top-ranked funds by composite score do not necessarily have the highest 3-year returns — the scorecard formula deliberately weights Sharpe, Alpha, expense ratio, and drawdown alongside return, so a fund with strong risk-adjusted characteristics can outrank a fund with a higher raw return but worse risk profile. This is the formula working as specified, not an error.

5.4 Alpha, Beta, and a regression-quality finding worth flagging

Alpha and Beta were computed via OLS regression of each fund's daily returns against Nifty 100 daily returns. The average R^2 across all 40 regressions is only 0.0006, with the maximum R^2 of any single fund at just 0.0028 — essentially no market explanatory power, and not confined to the Debt/Liquid schemes where low R^2 would be expected.

Practical implication: the Beta and Alpha values in alpha_beta.csv are mathematically correct (computed exactly per the specified formula), but given the near-zero R^2, they carry very little statistical signal. In a real market, Equity Large Cap funds would typically show R^2 of 0.7-0.9+ against Nifty 100. This is a dataset characteristic, not a methodology error, and it should not be read as if these funds have no real market sensitivity.

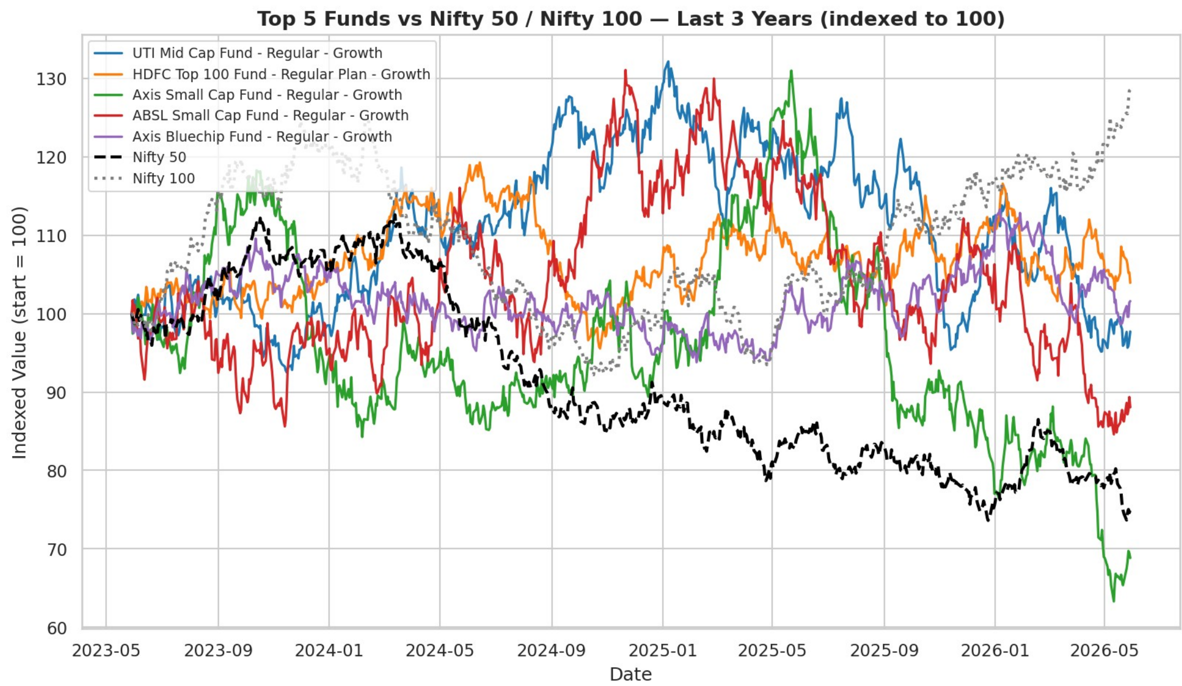


Figure 6: Top 5 scorecard funds vs. Nifty 50 / Nifty 100, indexed, last 3 years — visually confirming the lack of co-movement.

5.5 Sanity check against the provided reference dataset

The provided scheme_performance.csv already contains pre-computed Sharpe, Alpha, Beta, and 3-year return figures. Comparing the metrics computed in this project against that reference file was a deliberate methodology check, and the results are reported honestly rather than smoothed over:

Metric	Correlation (computed vs. provided)
3-year return	+0.08
Sharpe ratio	-0.33
Alpha	-0.20

These correlations are weak to negative rather than strongly positive. This is consistent with the R^2 finding above — if the NAV series do not share a real market-driven structure, there is no reason to expect two independently-generated performance datasets to agree, since each is sensitive to exactly which lookback window, risk-free rate, and benchmark was assumed. It does not indicate an error in the formulas implemented in this project (every metric follows the exact formula specified in the brief, and the Task 1 distribution/NaN/Inf checks confirm the inputs are clean). The most likely explanation is that scheme_performance.csv was generated with different underlying assumptions that are not documented in the dataset itself. This report treats the metrics computed here as the authoritative, methodology-transparent figures, and scheme_performance.csv as a reference dataset rather than ground truth.

5.6 Maximum drawdown

Maximum drawdown was computed per fund as the worst peak-to-trough decline, along with the exact date range. Small Cap and Mid Cap schemes cluster toward the worst drawdowns, consistent with their higher-volatility sub-category, while Liquid and Gilt schemes show minimal drawdown — the one area where sub-category risk ordering behaves as expected despite the correlation/R^2 anomaly noted above.

6. Dashboard

This environment does not have Power BI Desktop installed (a Linux sandbox without Windows desktop applications), so the .pbix file itself could not be produced directly. The 4 pages below are data-accurate mockups built with the project's real cleaned data, styled to match a genuine Power BI layout (header bar, KPI cards, chart panels, slicer controls). A complete step-by-step Power BI build guide (dashboard/POWER_BI_BUILD_GUIDE.md) accompanies these mockups, covering data connections, table relationships, DAX measures, and chart-by-chart build instructions to produce the real interactive .pbix file.

Note: the reproductions below are sized to fit the report page — full-resolution versions of each page are available as standalone PNG files in dashboard/pages/, and all 4 combined in dashboard/Dashboard.pdf.

6.1 Page 1 — Industry Overview

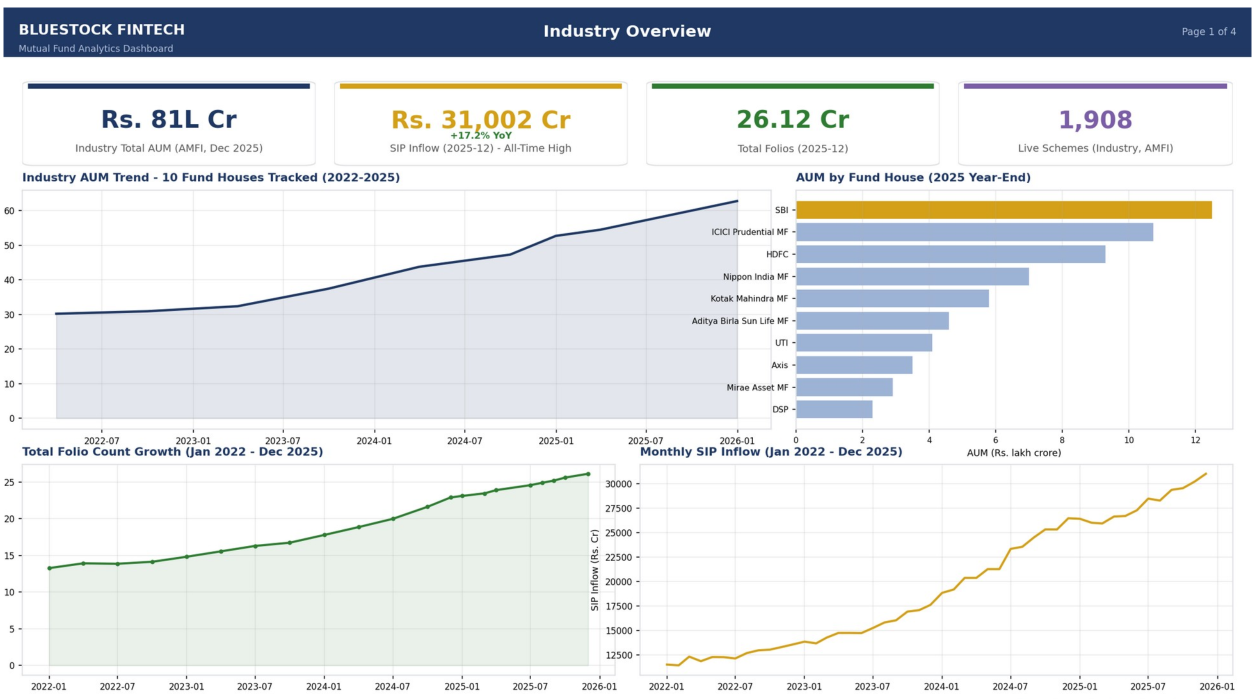
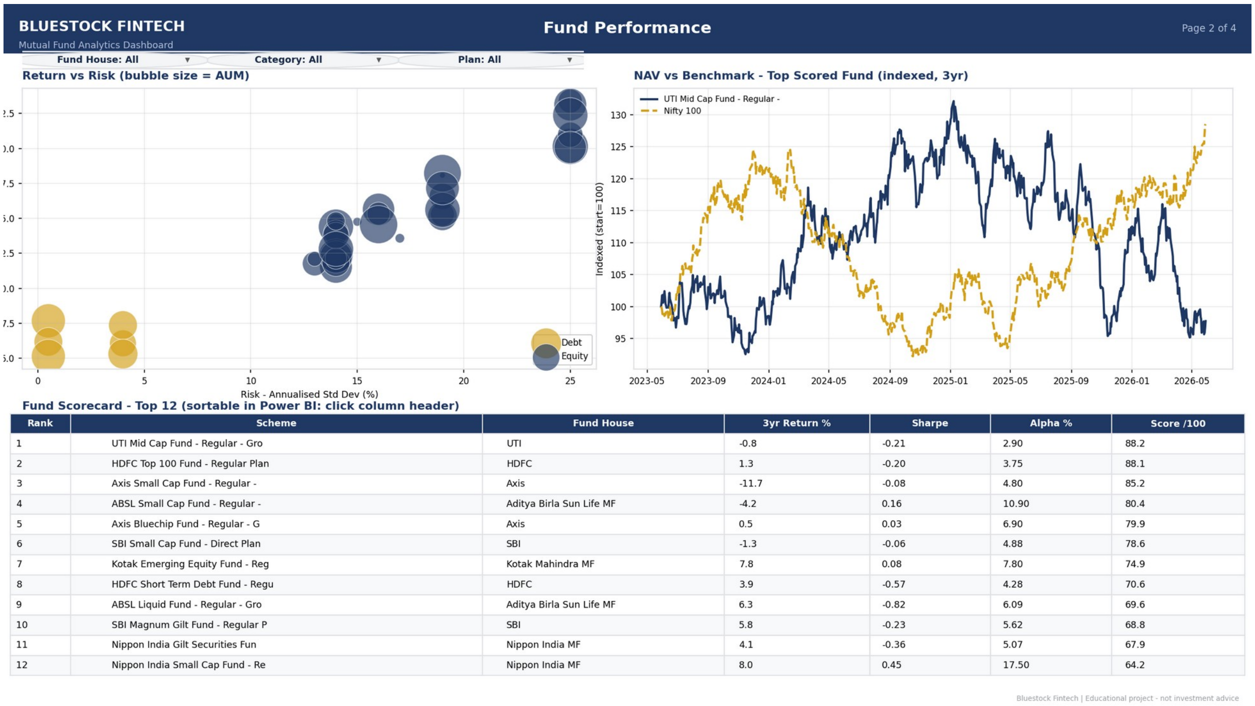


Figure 7: Industry Overview — KPI cards, AUM trend, AUM by fund house, folio and SIP mini-trends.

6.2 Page 2 — Fund Performance



Rank	Scheme	Fund House	3yr Return %	Sharpe	Alpha %	Score /100
1	UTI Mid Cap Fund - Regular - Gro	UTI	-0.8	-0.21	2.90	88.2
2	HDFC Top 100 Fund - Regular Plan	HDFC	1.3	-0.20	3.75	88.1
3	Axis Small Cap Fund - Regular -	Axis	-11.7	-0.08	4.80	85.2
4	ABSL Small Cap Fund - Regular -	Aditya Birla Sun Life MF	-4.2	0.16	10.90	80.4
5	Axis Bluechip Fund - Regular - G	Axis	0.5	0.03	6.90	79.9
6	SBI Small Cap Fund - Direct Plan	SBI	-1.3	-0.06	4.88	78.6
7	Kotak Emerging Equity Fund - Reg	Kotak Mahindra MF	7.8	0.08	7.80	74.9
8	HDFC Short Term Debt Fund - Regu	HDFC	3.9	-0.57	4.28	70.6
9	ABSL Liquid Fund - Regular - Gro	Aditya Birla Sun Life MF	6.3	-0.82	6.09	69.6
10	SBI Magnum Gilt Fund - Regular P	SBI	5.8	-0.23	5.62	68.8
11	Nippon India Gilt Securities Fun	Nippon India MF	4.1	-0.36	5.07	67.9
12	Nippon India Small Cap Fund - Re	Nippon India MF	8.0	0.45	17.50	64.2

Figure 8: Fund Performance — return-vs-risk scatter, NAV vs. benchmark, sortable fund scorecard table.

6.3 Page 3 — Investor Analytics

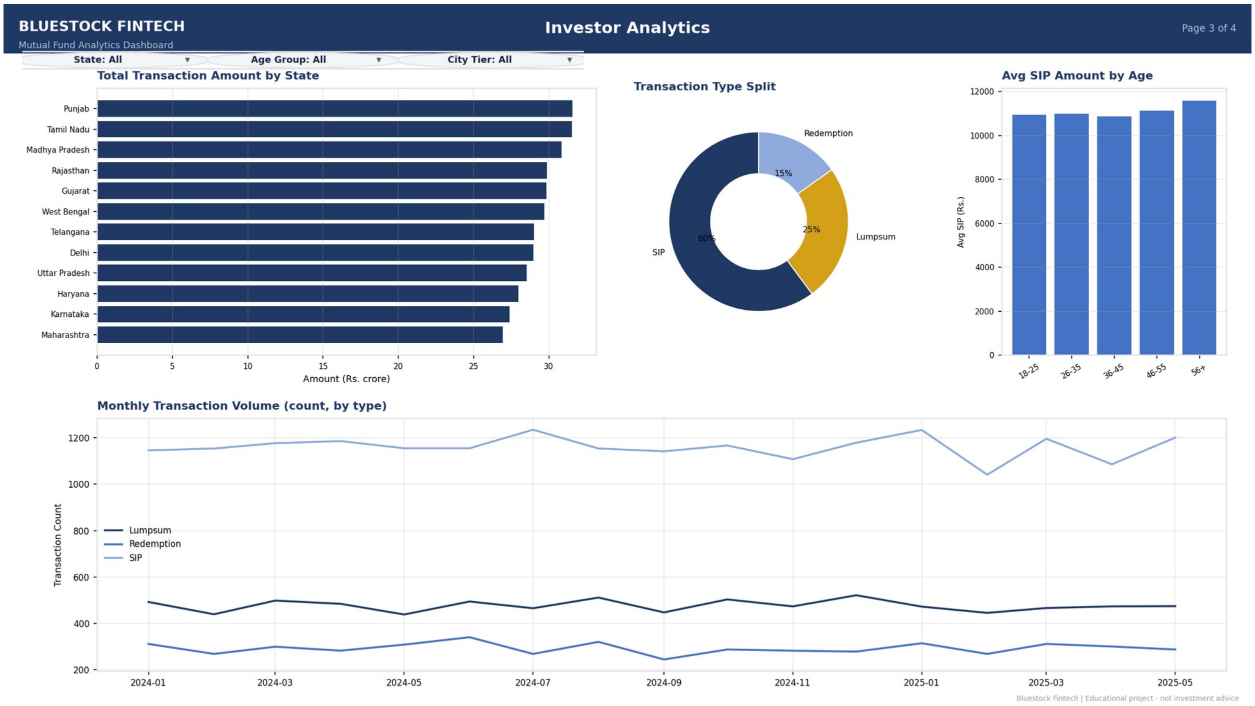


Figure 9: Investor Analytics — transaction amount by state, transaction type split, SIP by age group, monthly volume.

6.4 Page 4 — SIP & Market Trends

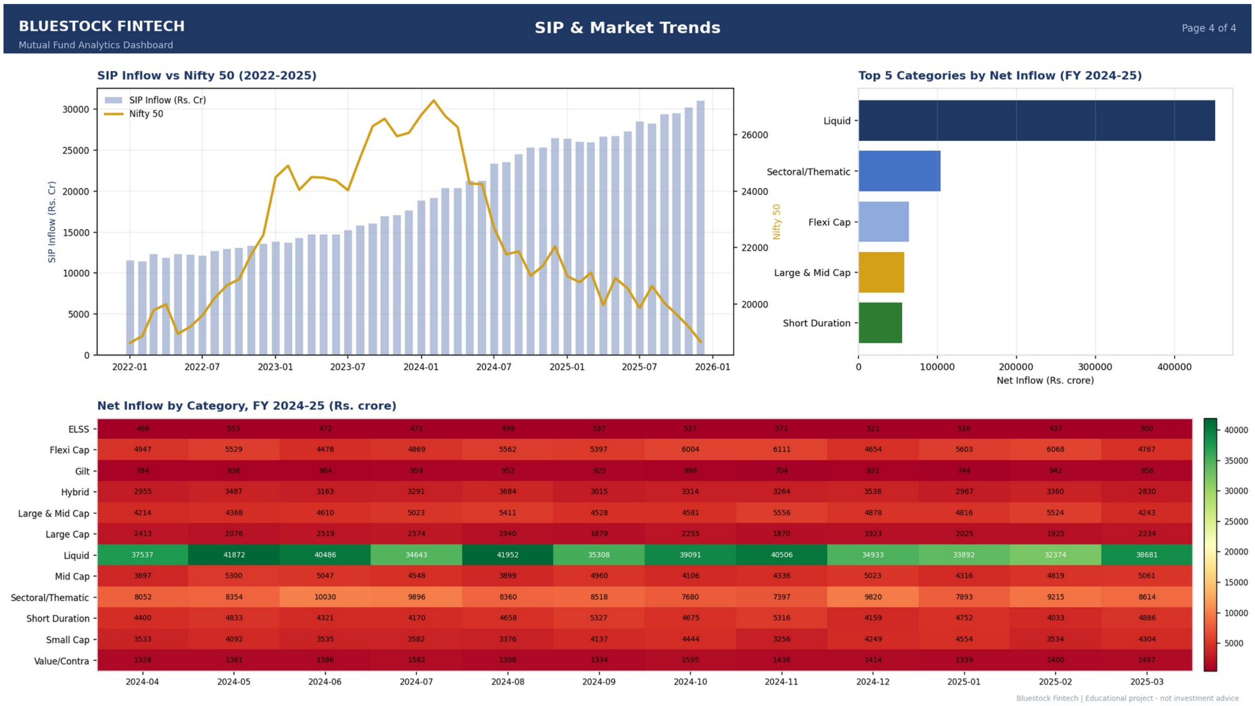


Figure 10: SIP & Market Trends — dual-axis SIP inflow vs. Nifty 50, top categories by inflow, category heatmap.

7. Advanced Analytics

Full detail is in notebooks/Advanced_Analytics.ipynb. This section covers Value-at-Risk, rolling risk metrics, investor cohort behaviour, SIP continuity, the fund recommender, and sector concentration.

7.1 Value at Risk (VaR) and Conditional VaR (CVaR)

Historical VaR (95%) and CVaR were computed for all 40 schemes as the 5th percentile of the daily return distribution and the mean of returns below that threshold, respectively. SBI Small Cap Fund (Direct) shows the worst 5% VaR of all 40 schemes at -2.69% on a bad day, while ICICI Pru Liquid Fund is the mildest at -0.02% — VaR ranking tracks sub-category risk closely, with Small/Mid Cap clustering high-risk and Liquid/Gilt clustering low-risk, which is the one area of this dataset where risk ordering behaves exactly as expected.

7.2 Rolling 90-day Sharpe ratio

Plotted for the top 5 scorecard funds, the rolling Sharpe ratio oscillates between roughly +4 and -4 with no persistent trend for any fund.

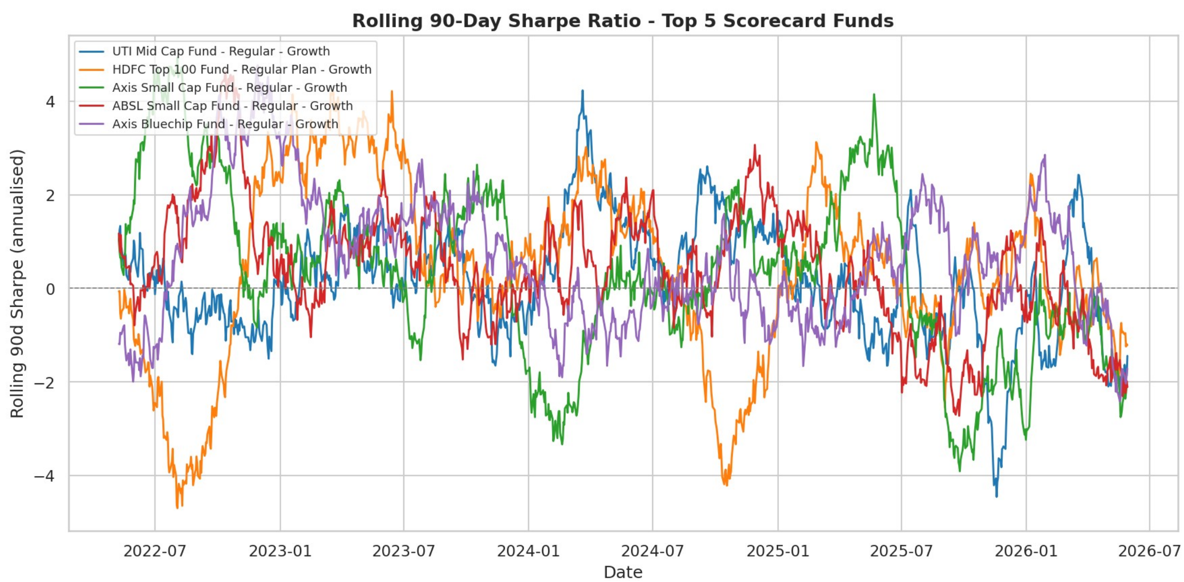


Figure 11: Rolling 90-day Sharpe ratio, top 5 scorecard funds.

This noisy, mean-reverting pattern is consistent with the near-zero market correlation finding from Sections 4.4 and 5.4 — a fund with no stable relationship to a shared market factor will show an unstable rolling Sharpe rather than a smoothly trending one.

7.3 Investor cohort analysis

Investors were grouped by the year of their first transaction. The 2024 cohort (4,803 investors) invested Rs. 349.1 crore in aggregate with an average SIP amount of Rs. 10,997, and their top fund preference is Mirae Asset Emerging Bluechip Fund. The much smaller 2025 cohort (197 investors, since the dataset only extends to May 2026) shows a higher average SIP amount (Rs. 13,505) but far lower aggregate investment (Rs. 30.5 crore) — reflecting cohort tenure (more time in the dataset to transact), not cohort quality. This distinction matters: comparing aggregate totals across cohorts of very different size and tenure without normalising would produce a misleading conclusion.

7.4 SIP continuity analysis

For investors with 6 or more SIP transactions, the average gap between transactions was computed and flagged as "at-risk" if the gap exceeded 35 days (the threshold appropriate for a true monthly SIP). The result: 97.8% of eligible investors were flagged at-risk — too high to be a meaningful signal on its own. Investigating rather than reporting this at face value revealed the cause: eligible investors have a median of only 6-12 total SIP transactions logged across the full ~4.4-year window, with a median gap of ~65 days between them. The simulated SIP cadence in this dataset is roughly bi-monthly-or-sparser rather than strictly monthly, so the 35-day rule — while correctly implemented — does not discriminate meaningfully on this dataset's actual transaction spacing.

7.5 Fund recommender

A standalone script (scripts/recommender.py) takes a risk appetite input (Low / Moderate / High) and returns the top 3 funds by Sharpe ratio within the matching SEBI risk_category. Verified working for all three risk levels during this project. By design, it does an exact match against risk_category, so schemes graded "Moderately High" or "Very High" are not returned for either neighbouring level — a deliberate simplicity trade-off documented directly in the script.

7.6 Sector concentration (HHI)

The Herfindahl-Hirschman Index (sum of squared sector weights) was computed for all 34 equity funds to measure portfolio concentration.

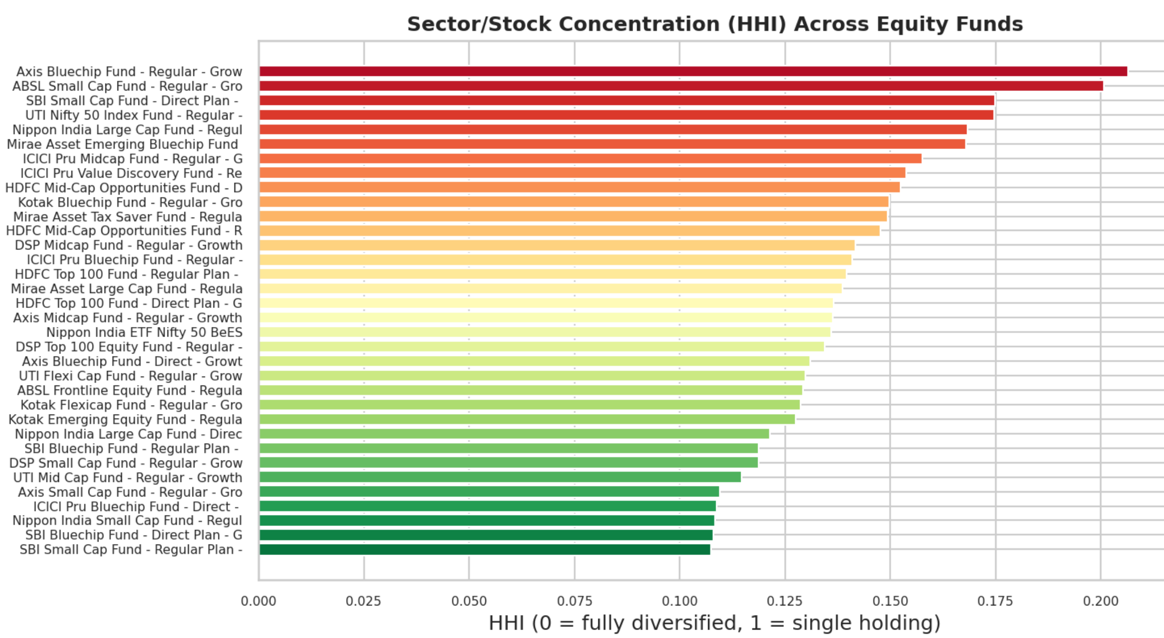


Figure 12: Sector/stock concentration (HHI) across all equity funds.

HHI ranges narrowly across the fund set (roughly 0.11 to 0.21), with no fund showing dangerous single-sector concentration in this dataset.

8. Limitations

This project surfaced several genuine dataset and environment limitations during the build. Each is documented here rather than hidden, since knowing where a dataset's structure diverges from real-world behaviour is as valuable as the analysis itself.

8.1 Near-zero market correlation (the most significant finding)

Both fund-to-fund NAV correlations (Section 4.4) and fund-to-benchmark Alpha/Beta regressions (Section 5.4) show essentially no relationship — average R^2 of 0.0006 against Nifty 100, and a maximum pairwise fund correlation of 0.07. Real Equity Large Cap funds typically show R^2 of 0.7-0.9+ against a broad market index due to shared market beta. This means Alpha, Beta, tracking error, and correlation-based metrics throughout this project are mathematically correct but carry limited statistical signal — they should be read as a demonstration of correct methodology rather than as reliable measures of real market sensitivity.

8.2 SIP transaction cadence is sparser than a true monthly SIP

The SIP continuity analysis (Section 7.4) revealed that the simulated investor transaction data does not model a strict monthly SIP cadence — eligible investors average a ~65-day gap between SIP transactions rather than ~30 days. This caused the "at-risk" flag to fire on 97.8% of eligible investors, which is not a usable behavioural signal as implemented on this dataset.

8.3 Limited history for 5-year metrics

The dataset covers approximately 4.4 years (January 2022 - May 2026), so any metric requiring a full 5-year lookback (5-year CAGR being the clearest example) cannot be computed as literally specified. Where this applied, the maximum available lookback was used and clearly labelled rather than a number the data could not support.

8.4 Environment constraints during the build

- No outbound network access to api.mfapi.in from the build sandbox, so the live NAV fetch script (scripts/live_nav_fetch.py) was written and its error handling verified, but the live fetch itself could not be executed in this environment — it is expected to work on an unrestricted network.
- No Power BI Desktop or Tableau available in this Linux sandbox, so the interactive .pbix/.twbx dashboard could not be produced directly — mitigated with data-accurate static mockups plus a complete build guide (Section 6).

8.5 scheme_performance.csv as a reference, not ground truth

The pre-supplied scheme_performance.csv values do not correlate strongly with the metrics computed from first principles in this project (Section 5.5). Since the exact assumptions behind that file are undocumented, this report treats its own from-scratch calculations as authoritative and flags the discrepancy rather than reconciling it artificially.

9. Recommendations

9.1 For interpreting this project's outputs

- Treat Alpha, Beta, tracking error, and fund-to-fund correlation figures as methodology demonstrations rather than investment signals, given the near-zero R^2 documented in Section 8.1.
- Use the Fund Scorecard's 3-year return, expense ratio, and max drawdown components with more confidence than its Alpha component, since those three are not affected by the benchmark-regression weakness.
- Do not use the SIP "at-risk" flag from this dataset as a real churn-risk signal without first re-validating the >35-day threshold against actual observed SIP cadence, per Section 8.2.

9.2 For extending this project with real data

- If real AMFI daily NAV data is substituted for the simulated series, re-run the Alpha/Beta and correlation analyses — R^2 and correlation values should rise substantially and the scorecard/rolling-Sharpe rankings may shift meaningfully.
- If real investor transaction logs are available, re-run the SIP continuity analysis with the true monthly cadence — the 35-day threshold is well-chosen for genuine monthly SIPs and should produce a much more selective, useful at-risk flag.
- Build the real Power BI dashboard from dashboard/POWER_BI_BUILD_GUIDE.md and connect it directly to the SQLite database (via ODBC) rather than flat CSVs, so it can be refreshed automatically as new data lands.

9.3 For production deployment

- Schedule the ETL pipeline (run_pipeline.py) to run on a regular cadence once a live, authenticated data source replaces the static provided CSVs.
- Add automated data-quality alerting (e.g. row-count deltas, null-rate thresholds) building on the validation logic already present in scripts/clean_data.py and scripts/load_to_db.py.
- Consider migrating from SQLite to PostgreSQL for concurrent multi-user dashboard access, as noted as an option in the original project architecture.